

Progress Report: Smart Veterinary System with AI-Based Disease Detection

Course Code: CSE 299

Section: 04

Project Group: 05

Group Members:

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Abstract—This progress report summarizes the development work completed this week for the project titled *Smart Veterinary System with AI-Based Disease Detection*. The team successfully transitioned into the core AI phase, focusing on data collection, machine learning model training in Google Colab, and integrating the trained model into the Django backend. Furthermore, a dynamic AI assistant powered by the Gemini API was implemented. Tasks were distributed equally, with each member contributing to the successful integration of AI components into the web platform.

I. CURRENT PROGRESS

During this week's development session, the team achieved major milestones regarding the project's artificial intelligence capabilities. The focus shifted from standard web features to data processing, model training, and dynamic API integration.

TABLE I
THIS WEEK'S DEVELOPMENT TASKS

Completed Task	Responsible Member
Data collection, dataset preparation, and training the initial ML model in Google Colab	Nafia Haque Muskan
Integrated the Gemini API to dynamically support and power the AI assistant page	Rishat Bin Sultan
Implemented the trained ML model into the Django backend for real-time predictions	Fariha Islam
Smoothed the AI assistant frontend dynamically and ensured seamless user interaction	MD Tamim Hasan

II. CHALLENGES AND DECISIONS

Challenges:

- Exporting the trained model from Google Colab and ensuring compatibility within the local Django environment.
- Managing asynchronous requests and handling latency when calling the external Gemini API for the assistant.
- Dynamically updating the frontend UI to display real-time AI responses without breaking the user experience.

Decisions:

- The team utilized the Gemini API as a supplementary intelligent layer to support the primary, custom-trained disease detection model.

- Decided to lock further feature development on the web application to dedicate all remaining time to refining the AI models and system stability.

III. FUTURE PLAN

With the core AI systems and the web platform successfully integrated, the next and final phase of development will focus heavily on quality assurance and optimization.

- Conduct comprehensive system testing across all modules to identify and resolve existing bugs.
- Smooth and optimize the machine learning model to improve prediction accuracy and reduce false positives.
- Refine the Gemini API prompts to ensure the AI assistant provides highly accurate, context-aware veterinary advice.
- Perform end-to-end integration testing to ensure the frontend, Django backend, and AI models communicate flawlessly.

IV. CONCLUSION

This week marked the most critical phase of the project, successfully bringing the "AI" into the *Smart Veterinary System*. The team effectively collaborated to gather data, train a functional ML model in Colab, and deploy it within the Django architecture. The addition of a dynamically smoothed AI assistant, backed by the Gemini API, significantly enhances the platform's utility. The upcoming week will be dedicated strictly to testing, bug fixing, and model optimization to finalize a robust, production-ready system.